

Checkout Chaos

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Checkout Chaos is a 3D shopping-cart racer in Godot 4. One human and three bot shoppers compete at a grocery store’s grand-opening sale. The browser prototype uses Three.js; the game uses a third-person chase camera, low-poly shapes and bright flat colors.

1 Core loop

Section author: Rickey

Grab items, protect or steal a haul by ramming rival carts, and drive out the front doors to check out before time runs out, then return for another haul.

2 The four systems

Section author: John

The game has four separate systems, one per teammate: Player (Rickey), Cart (Evan), Rivals (John), and Store / Round Manager (Anthony). Player and Rivals independently produce the same DriveCommand interface for Cart.

System and owner	State it owns	What it does
Player Owner: Rickey	Human input, camera, HUD, Shopper ID screens and feedback state.	PlayerController produces DriveCommand; handles controls, camera, UI, popups, sound and rumble.
Cart Owner: Evan	Position, velocity and speed; boost meter; items[] and value; stun and immunity timers.	Moves all carts, enforces the 24-item cap and weight slowdown, resolves ram-steal, and transfers or clears carried inventory.
Rivals Owner: John	Bot targets, paths, decision timers, greed, aggression, boost habit and unsticking state.	BotController produces DriveCommand; bots choose pickups, ram, bank, reroute and recover from being stuck.
Store / Round Manager Owner: Anthony	Round phase and timer; doors; world pickups and spawns; hazards; banked scores, stamps and standings.	Runs the store, Deal of the Day, checkout and round flow; creates spills and records receipts and match results.

Each state has one writer. Player reads gameplay state for display; Rivals reads it for bot decisions. Store requests inventory changes through Cart; it does not edit items[] directly. Cart reports completed transfers to Store.

3 The seams

Section author: Evan

The following named contracts cross system boundaries. Writer means the producer of the message or state; reader means its consumer. Player and Rivals are separate systems with a shared command format.

Name	What crosses	Writer	Reader
DriveCommand	throttle, brake, steer (-1 to 1), boost; one command per cart per update.	Player or Rivals	Cart
ItemData	category, value, mesh, is_deal; the item data used by pickups, inventories and targeting.	Store / Round Manager	Cart; Rivals
CartState	speed, items[], value, is_stunned; read-only snapshot of cart state.	Cart	Store; Rivals; Player (HUD)
cart_robbed(winner, loser, items, spilled)	Winning and losing carts, transferred items and overflow items. Emitted after Cart resolves the collision and inventory transfer.	Cart	Store (spawn spills); Player (popup); Rivals (retarget)
item_collected(cart, item)	Accepted pickup and receiving cart. Store removes the world pickup once Cart accepts it.	Cart	Store; Player (sound)
checked_out(cart, value)	Cart and successfully banked value, after inventory is cleared.	Store / Round Manager	Player (popup and scoreboard)
round_started(round) and round_ended(results)	Round number or completed results; controls become active or stop.	Store / Round Manager	Player; Rivals
hazard_spawned(hazard)	New hazard and affected location.	Store / Round Manager	Rivals (reroute)

Checkout and event ownership

Store detects entry into the checkout zone and requests that Cart return and clear its current haul. Store banks the returned value exactly once and then emits checked_out. This keeps Cart responsible for carried items and Store responsible for money already checked out.

Use cart_robbed(winner, loser, items, spilled) consistently. The items field means items actually transferred to the winner; spilled means overflow for Store to spawn. Each physical item must end up in exactly one cart, on the floor, or checked out.

Integration

Each owner builds a test scene with fake inputs. Integrate Cart + Player first, then Store for the round loop, then Rivals. The Store owner edits Main.tscn at merges. PlayerController and BotController both drive Cart.tscn through DriveCommand.

4 Our reading of Inheritance

Section author: Anthony

We read Inheritance as receiving someone else's accumulated wealth and legacy. Grandma's Card expresses the family legacy directly; ram-steal and spilled items make possession pass between shoppers during play.

Grandma's Card

Grandma was the store's most famous shopper. She leaves you her Platinum Shopper ID card and one instruction: "Don't let Carl win." Her old rivals and their grandkids compete against you. Each round is one day of the sale weekend; a round win adds a stamp, and winning the match reissues the card in your name.

The Shopper ID card appears on the title, character-select and results screens. Its front has the store logo, name, photo, member number, "Member since" year and tier badge. Its back has a barcode, lifetime savings and round-win stamps. Each bot has a card for its introduction and personality.

Rival	Tier	Style and personality
Coupon Carl	Gold	Rammer; clips every coupon and will run you over for a deal.
Aunt Bev	Platinum	Safe banker; checks out early and often.
Rolling Rita	Silver	Speed demon; boosts everywhere and holds out for a giant haul.

Ram steal and spilled items

When two carts touch, compare their speeds. The faster cart wins only if it is moving at least 5 m/s and at least 1.5 m/s faster than the other cart. Otherwise both bounce and nothing changes hands. The loser loses its entire haul, is knocked back, is stunned for 0.7 s and receives 1.6 s of immunity against another steal.

The winner receives the haul up to its 24-item cap. Overflow spills onto the floor as normal pickups anyone can collect, including the loser. A spilled Deal of the Day remains gold and worth \$100. Parking and carrying a heavy haul create risk; a well-timed hit can enable a comeback.

Example: Rita has 20 items worth \$260 and you have 8 worth \$110. If Rita wins the collision, four of your items fit and four spill. If the transferred four are worth \$50, Rita now carries 24 items worth \$310, you carry none, and the floor holds \$60. Total item value remains \$370.

Where the theme lives

The story and Shopper ID screens show inherited identity and rivalry. Cart resolves the transfer of wealth, Store turns overflow into claimable pickups, Player presents the event, and Rivals lets bots react. Ram-steal is a playful mechanical interpretation; Grandma's Card supplies the literal family inheritance. Carts reset empty each round; no inherited stat perk is included.

5 Art direction

Section author: Evan

One shared art direction applies to the whole team and to all four systems.

Field	Team direction
Medium and visual style	Godot 4, 3D; low-poly shapes, bright flat colors, no textures except signage.
Camera and perspective	Third-person chase camera about 8.5 m behind and 5.5 m above the cart, looking ahead. FOV widens from 62° to 72° on boost; brief shake on impact.
Palette and value cues	Produce green; bakery orange; dairy white; snacks red; frozen blue; electronics purple. Deal of the Day is gold with a light beam. Pair colors with aisle signs, item shapes and displayed values.
Environment	Six grocery aisles; checkered mint-and-cream linoleum; hanging aisle signs; red-and-yellow grand-opening banner; front-door checkout zone.
Shopper identification	Colored cart rims, handles and flags, plus floating name tags. Player is yellow. Shopper ID cards carry the characters' identity and tier.
HUD layout	Timer and round at top left, minimap below; banked totals and current cart values at top right; item count / 24, cart value and boost at bottom center; steal and checkout feed at bottom left.
Menus and results	Shopper ID on title, character-select and results screens. Results look like a store receipt and show the card, stamps and standings.
Animation and feedback	Items fly between carts on a steal; doors animate open and shut; camera shakes on rams; falling displays wobble 1 s before collapse. Popups report steals and checkouts.
Music and sound	Upbeat supermarket muzak speeds up during the final 20 s. PA calls opening, Deal of the Day, hazards and final call. Pickup blip, crash thud, steal whoosh, register ding and squeaky wheel loop.

Controls supporting the presentation

Keyboard: W / Up for gas; S / Down for brake and reverse; A D / Left Right to steer; Shift / Space for boost; Esc to pause. Gamepad: right trigger, left trigger, left stick, A / Cross and Start respectively. Player feedback includes sound, popups and controller rumble.

6 Scope

Section author: John

What is in

One store, one human, three bots and the four systems in Section 2. Keep the arcade cart, chase camera, 24-item inventory, slowdown, boost, ram-steal, overflow pickups, Shopper ID screens and Grandma’s Card introduction.

Cart targets about 15 m/s top speed. Each item reduces speed by 1.2% (about 29% at capacity). Boost adds 8 m/s to top speed, drains in about 2 s and refills in about 8 s. Rivals decide every 0.3 s, use NavigationAgent3D on a baked navmesh and reverse and turn when stuck for 1 s. Greed, aggression and boost habit distinguish the three bots.

Aisle or item	Value	Spawn weight
Produce	\$5	High
Bakery	\$10	High
Dairy	\$10	High
Snacks	\$15	Medium
Frozen	\$20	Low
Electronics	\$40	Rare
Deal of the Day	\$100	One at a time; every 14–22 s

Store supports up to 46 floor items and a regular spawn every 0.5 s. Included hazards are wet-floor spills (no steering for 1 s, yellow warning sign), a crossing pallet jack (blocks and bumps carts), and falling displays (1 s warning, 5 s blockage).

Match rules

Play three 2-minute rounds, with a 3 s countdown and a 10 s receipt/results screen between rounds. Final call is the last 20 s. Only checked-out value counts. A round win earns a stamp; most stamps wins the match, with total banked money across all three rounds breaking a stamp tie. Exact remaining ties share the result. Empty carts reset each round; hazards and bot aggression increase without shortening the timer.

Three cuts from this build

Cut	What is excluded and why
1 Multiplayer system	An additional fifth system for network sessions, authority and synchronization, plus local split-screen. Keep one human and three bots so networking and multiple views do not expand this build.
2 Cart types	Heavy, fast and big-basket variants. Keep the shared Cart scene, 24-item cap and common movement rules to avoid balancing separate vehicle classes.

Cut	What is excluded and why
3 Extra hazards	Sample lady and Cleanup on aisle X closures. Keep the three core hazards so extra slowdowns and dynamic aisle rerouting do not expand the Store and Rivals work.

7 Which seam breaks first and why

Section author: Rickey

We predict the Cart → Store / Round Manager seam, `cart_robbed(winner, loser, items, spilled)`, breaks first.

Cart decides who wins and removes or transfers the haul, while Store creates the overflow pickups. A duplicate collision callback, a different interpretation of items versus spilled, or checkout in the same update can duplicate items, lose value or bank a haul twice. The seam joins physics timing, inventory ownership and world spawning, so it is more fragile than the simple DriveCommand input contract.

The first integration check

Drive a 20-item cart into an 8-item cart at qualifying speeds. The winner must finish with 24 items, the loser with 0, and Store must create exactly 4 spilled pickups. Check item identities and total value, not just counts. Repeat the contact during immunity: no second transfer or extra spills should occur.

Then run the same collision at the checkout boundary and at round close. Verify that each item reaches exactly one destination and that checked-out value is recorded once. Include a \$100 Deal of the Day in the spill to verify its value and gold appearance survive.

Contract used to contain the risk

Cart resolves each steal once, updates both inventories and then emits one `cart_robbed` message. Store consumes that message once and spawns only spilled. Store processes checkout after pending steal transfers, through Cart's inventory-clearing operation. At round close, finish already accepted transfers and checkout before finalizing scores; reject new gameplay actions after the phase closes.

If value is lost or duplicated, first log the collision, the transferred and spilled item identities, and the checkout event in order. Test Cart with a fake Store listener, then reconnect Store. This directly tests the boundary where ownership changes.

Other implementation risks

Bot pathing can fall back to hand-placed aisle-end waypoints from the prototype. Jittery cart collisions can fall back to kinematic carts with the same speed rule. Shared-scene conflicts are limited by assigning `Main.tscn` merges to the Store owner.